



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Programming and Problem Solving

Course Code: TE7286

Faculty: Engineering

Course Credit: 2

Course Level: 1

Sub-Committee (Specialization): Computer Science

Learning Objectives:

The students are able to:

Understand and develop Computational Thinking concepts

Understand the programming in IDE (Integrated Development Environment) and write, execute and debug simple programs.

Understand basic Python Programming, functions, patterns, algorithmic constructs etc.

Understand and apply object oriented programming concepts

Books Recommended:

Book	Author	Publisher
Computational Thinking: The Beginners Guide To Problem solving & Programming	Karl Beecher	BCS, The Chartered Institute for IT (11 August 2017).
Python The Complete Reference	Martin C Brown	McGraw Hills

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Introduction of Problem Solving: Objectives and outcomes of Computational Thinking How is Computational Thinking used Concepts of logical Thinking and Algorithms Flow charts Objectives of Problem Solving Defining a problem devising solutions. Conceptual introduction: topics in computer science algorithms modern computer systems: hardware architecture data representation in computers software and operating system	6	6
2	Basics of Python: Python Interface when to use Python comments literal constants numbers variable identifier naming data types strings basic I/O statements	6	6

	logical and physical line indentation operators and expressions tuples dictionary lists: indexing slicing basic methods control flow: conditionals and iterators		
3	Functions and Modules: Function parameters local statement global statement default argument values keyword arguments VarArgs parameters return statement recursive functions Docstrings byte compiled .pyc files the from..import statement module's name making your own module dir function packages	8	8
4	Classes and Objects Defining your own classes superclasses subclasses inheritance searching for missing class components creating objects class attributes: class variables and instance variables defining adding and removing attributes explicit constructor invocation class methods: defining and using the self parameter meaning and usage inheritance and overriding finding class/object components single inheritance vs. multiple inheritance	10	10
Total		30	30

Evaluation:

Assignment
Examination

Pedagogy:

Class room teaching
Practical Assignments